



An Axiomatic Concurrency Model





PO

po



PO

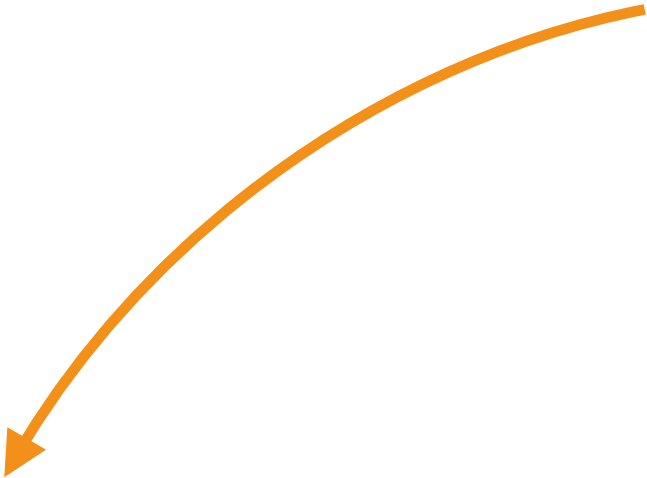
p

o

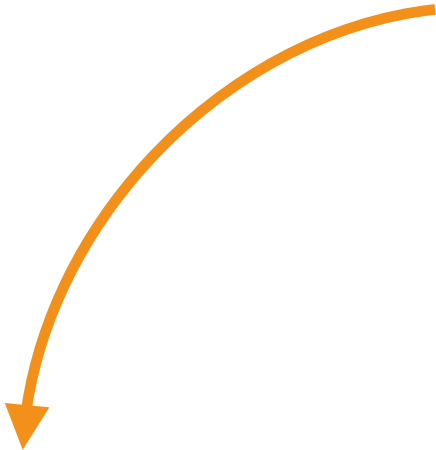


po

po



cosrc



collog

$\mathbb{E} = \text{Spawn} \cup \text{Run} \cup \text{Complete}$

$\text{comp} \subseteq \{\text{Cown} \times \text{Complete} \times \text{Run}\}$

$$\text{po} \subseteq \{\text{Run} \times \text{Spawn}\} \cup \{\text{Run} \times \text{Complete}\} \cup \\ \{\text{Spawn} \times \text{Spawn}\} \cup \{\text{Spawn} \times \text{Complete}\}$$



S



-

R  +

R



-



S



—



C



-

R  +

R  -



C



—

```
update(src, +100)
```

```
update(src, -100)
```

Candidate executions are tuples of (E, po, co)

```
def update(account: cown[Account], amount: int) {  
  when(account) { A  
    if(account.balance + amount > 0) {  
      account.balance += amount  
      val id = account.id()  
      when(log) { B log.record(id, amount) }  
    } } }  
}
```

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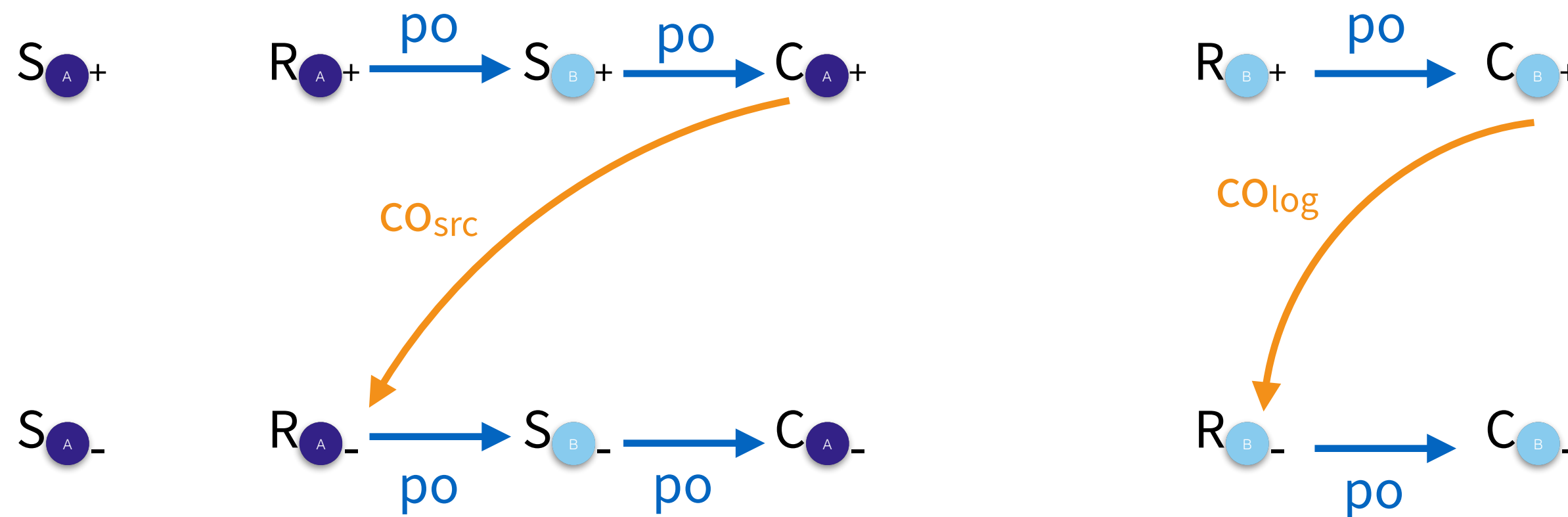
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Candidate executions are tuples of $(\mathbb{E}, po, co)$



$\mathbb{E} = \text{Spawn} \cup \text{Run} \cup \text{Complete}$

$po \subseteq \{\text{Run} \times \text{Spawn}\} \cup \{\text{Run} \times \text{Complete}\} \cup$
 $\{\text{Spawn} \times \text{Spawn}\} \cup \{\text{Spawn} \times \text{Complete}\}$

$co \subseteq \{\text{Cown} \times \text{Complete} \times \text{Run}\}$

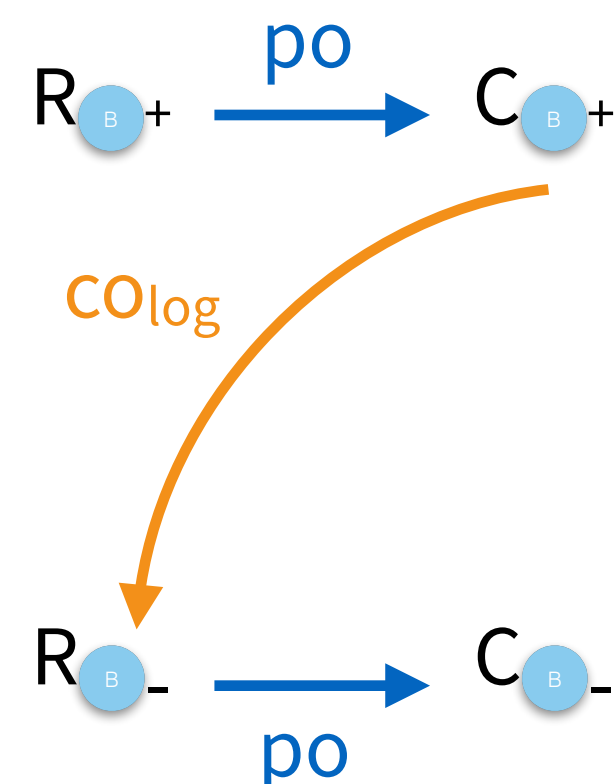
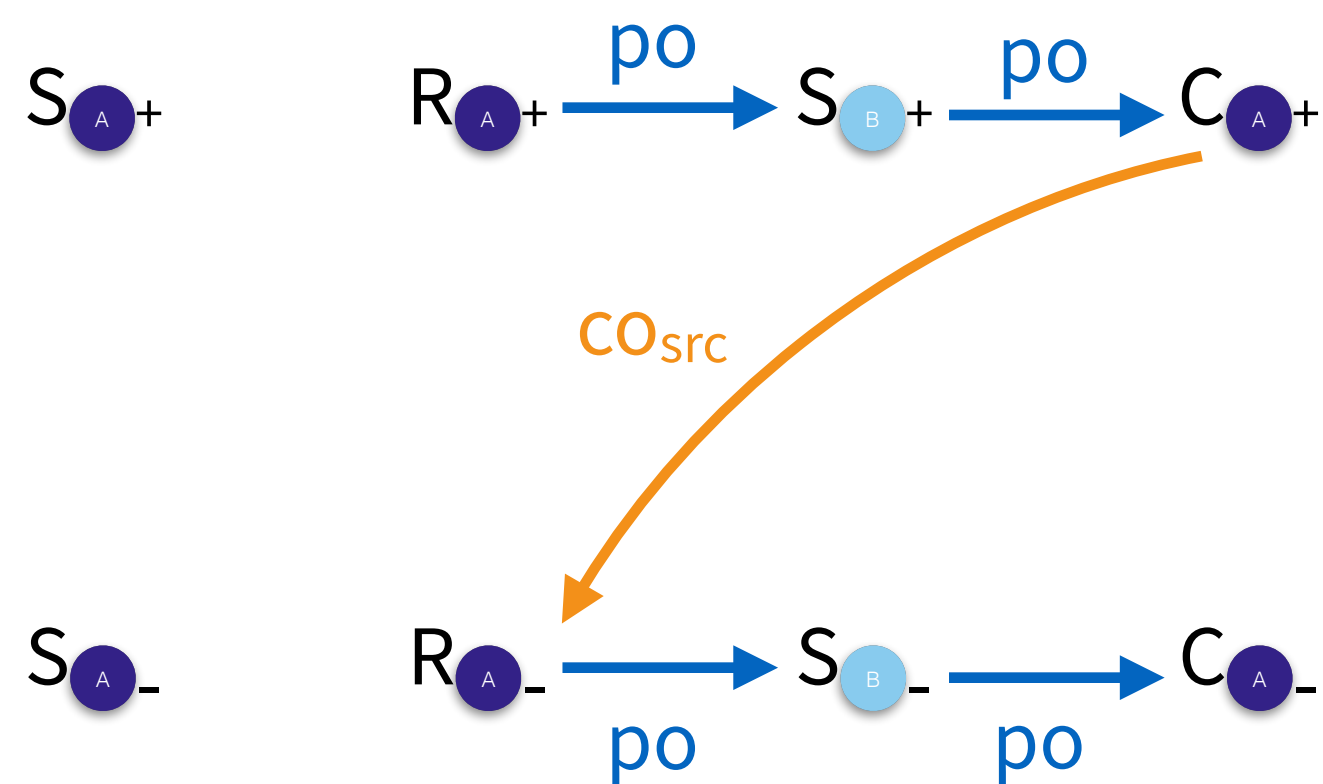
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We derive the run relation from behaviour spawn to run